THOMAS MORGAN

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PROFESSIONAL SUMMARY

I'm a recent graduate from the University of Birmingham with a strong interest in machine learning systems and data engineering. I started my career working on small-scale data projects and have developed practical experience with Python, SQL, and cloud platforms. I'm particularly drawn to problems involving data pipeline optimization and I'm keen to expand my understanding of modern ML infrastructure while contributing meaningfully to research-focused work.

WORK EXPERIENCE

Research Data Engineer

TechVenture Labs

techventure.io

Birmingham, United Kingdom

September 2023 - Present

Designed and built data collection pipelines for machine learning training datasets using Python and PostgreSQL

Implemented automated data validation checks that reduced downstream model training errors by 23 percent

Collaborated with three senior engineers to document data lineage and metadata standards for internal use

Conducted exploratory analysis on time-series data to identify patterns for feature engineering in predictive models

Junior Data Analyst

DataStream Solutions

datastream.co.net

Coventry, United Kingdom

June 2022 - August 2023

Developed SQL queries to extract and transform raw data from multiple source systems for business intelligence reporting

Built dashboard visualizations in Tableau to track KPIs for marketing and operations teams across five departments

Created Python scripts to automate weekly data reconciliation processes that previously required manual effort

Implemented version control workflows using Git to manage analysis code and improve team collaboration practices

Data Processing Intern

Innovate Research Group

innovate-research.io

Birmingham, United Kingdom

March 2021 - May 2022

Wrote Python scripts to clean and preprocess datasets containing over 2 million records for statistical analysis

Assisted in setting up LangChain workflows for natural language processing tasks in production systems

Documented data processing methodologies and created technical guides for team onboarding

Performed quality assurance testing on data pipelines and reported issues through JIRA ticketing system

EDUCATION

Bachelor of Science in Computer Science

University of Birmingham

Birmingham, United Kingdom

Graduated: June 2021

Relevant Coursework: Database Systems, Data Structures and Algorithms, Software Engineering, Statistics and Probability, Web Technologies

TECHNICAL SKILLS

Programming Languages: Python, SQL, JavaScript, Bash

Data Tools: Pandas, NumPy, Scikit-learn, PostgreSQL, MySQL

Cloud Platforms: AWS (S3, EC2), Google Cloud Platform

Orchestration and Infrastructure: Kubernetes, Terraform, ArgoCD

Version Control: Git, GitHub

Data Visualization: Tableau, Matplotlib, Seaborn

Other Tools: Jupyter Notebooks, Apache Airflow, Linux

PUBLIC REPOSITORIES

github.com/tmorgan-research/weather-data-pipeline - Python ETL pipeline for weather API data ingestion and storage

github.com/tmorgan-research/customer-segmentation - Machine learning clustering analysis using customer transaction data

github.com/tmorgan-research/sql-performance-analysis - Query optimization case studies and benchmarking tools

github.com/tmorgan-research/data-quality-framework - Generic data validation library for pipeline quality checks

CERTIFICATIONS

AWS Certified Cloud Practitioner (2023)

Google Cloud Associate Cloud Engineer (In Progress)